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CODE Yudh

PROJECT NAME

Nazar

TAGLINE

Stop the scam before the money moves

TEAM NAME

CodeVators

TRACK

Open Innovation

LIVE PROTOTYPE

nazar-codevators.vercel.app

ORGANISED BY

Department of Computer Science & Engineering

Chandigarh University



Team Introduction & Contributions

Team Members	Roles	Skills	Division of Responsibilities
Kunal Mittal	Team lead, backend and AI/ML	Backend, AI/ML, web and mobile development, DSA	Live audio pipeline, speech to text, scam detector
Ansh Kumar	Frontend and demo	React, TypeScript, UI design, demo building	Live screen, warning overlay, voice prompt, Telegram alert
Shivani Kumari	Product, data and pitch	Research, call scripting, testing, presentation	Call scripts, testing, business story, pitch

Problem Statement

- **Problem Definition**

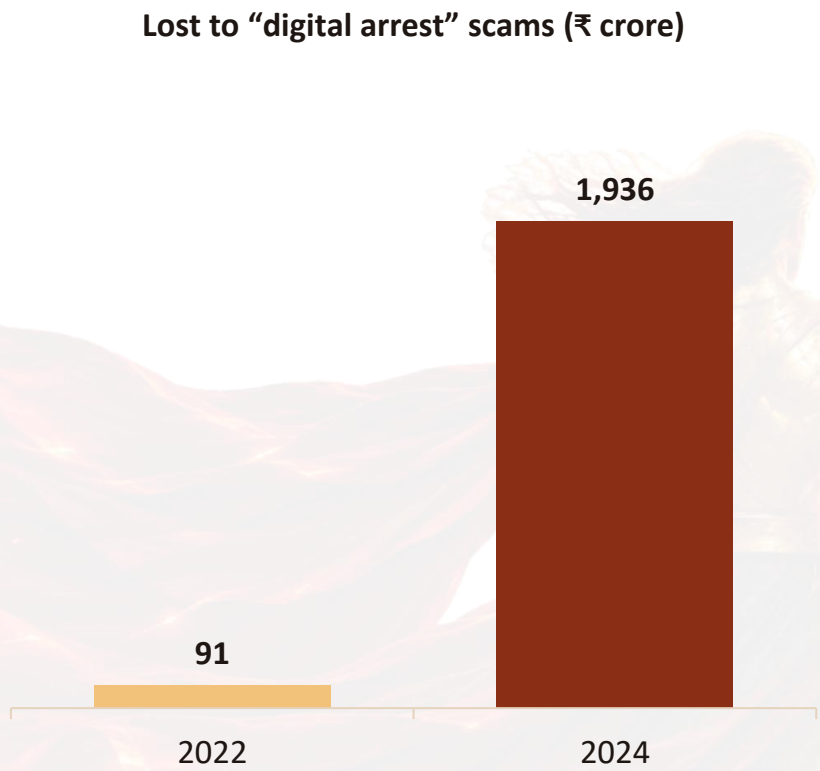
  Scammers call a victim, pretend to be an officer or bank staff, and pressure them to share OTPs or send money. The victim finds out only after the money is gone. Today's protection mostly works before or after the call, not during it.

- **Target Audience**

  Elders and first-time digital users, and the families who worry about them. Later: banks, payment apps and telecom operators who want to protect their customers.

- **Importance of the Problem**

  About ₹1,935 crore was lost to “digital arrest” scams in 2024 [2][3]. Only about ₹60 crore was refunded out of ₹36,448 crore lost to cyber fraud from 2021 to Feb 2025 [1]. Money is rarely returned, so the call itself must be stopped.



Sources: [1] Scroll.in, [2] Inc42, [3] Moneylife, citing figures given to Parliament in March 2025. Chart shows the two years reported in [3].

Problem Context & Existing Solutions

- **Background**
  In a “digital arrest” scam the caller poses as CBI, police or a bank, says you are in a crime case, keeps you on a video call, asks for OTP and details, then asks for money to a “safe account” [4][5][6].
- **Research**
  We studied I4C and state cyber-crime advisories [4][5][6] and the figures given to Parliament [1][2][3]. From them we built our own five-step detection framework (right). That each step uses words that can be spotted in speech is our hypothesis, and we will test it.
- **Current Solutions**
  Spam-number apps, bank SMS alerts and awareness campaigns.
- **Identified Gaps**
  Spam lists need a number to be reported first. SMS alerts come after the money moves. Campaigns rely on the user to remember. In our assessment, none of them acts live during the call, in Hindi or Punjabi, or alerts the family. That is the gap Nazar fills.

Our five-step detection framework



Sources: [1] Scroll.in, [2] Inc42, [3] Moneylife, [4] PTI/Tribune on I4C advisory, [5] Gujarat Cyber Crime, [6] CyberSecure India. Five-step framework: our synthesis of [4][5][6]. Gap analysis: our own.

Proposed Solution

Solution Overview

Nazar listens to a phone call on the user's own device, turns speech into text, scores the scam risk from 0 to 100, and warns the user the moment the risk is high.

Core Idea

Our five-step detection framework, drawn from cyber-crime advisories, models how these calls usually unfold. Nazar tracks how far a call has gone through it and acts before step 5, the money transfer.

How It Addresses the Problem

A live warning in Hindi, Punjabi or Hinglish, spoken aloud, plus an alert to a trusted family member. Audio stays on the device and is not saved.

Who gets what

Who	Value
Users and elders	A clear warning in real time, in their own language
Family	An instant alert, so they can call and stop the transfer
Banks and telecom partners	Fewer fraud cases and a protection feature to offer (expected, to be tested)

Business model: free for families; partners pay a licence or per-user fee. Not yet validated.

Key Features & Innovation

- **Core Features**

  Live speech to text; a five-step risk score with reasons; a full-screen and spoken warning; a Telegram alert to family; a demo mode with scripted calls.

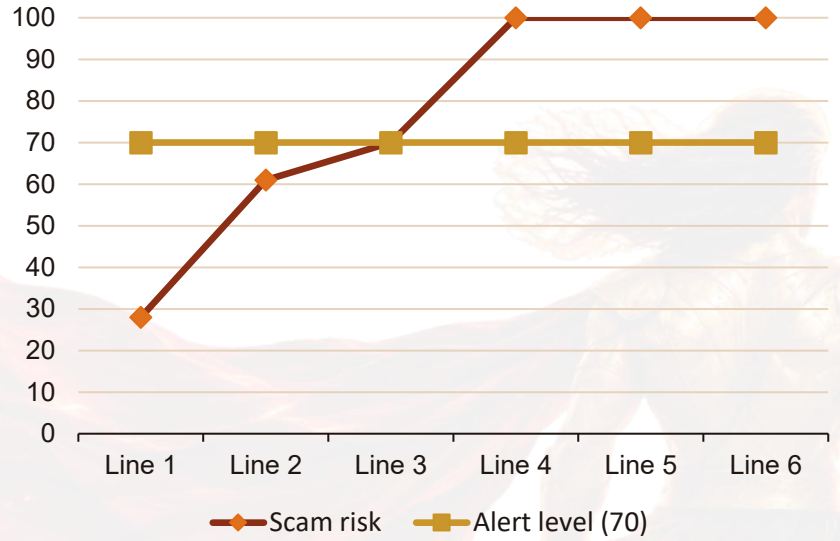
- **Unique Selling Points**

  Works during the call. Works on new numbers because it listens to what is said. Hindi, Punjabi and Hinglish. Family in the loop. Free for families, licensable by banks and telecom operators.

- **Innovative Aspects**

  It scores the progress of the scam script, not single words. A genuine line like “never share your OTP” lowers the score, so real bank calls are not flagged. Privacy by design: on-device, no audio stored.

Risk score in the prototype, sample call S1



Starting weights: Authority 0.35, Fear 0.50, Isolation 0.70, Control 0.80, Transfer 0.90. Alert at 70 or more. Real output of our prototype on scripted call S1: the alert fires at line 3, three lines before money is asked for.

Later we plan to add a MuRIL-based classifier, which understands Hindi, Punjabi and Roman-script Hindi [9].

Proposed Workflow

- **Working Mechanism**

  Microphone audio goes through a voice filter and Whisper large-v3-turbo on the device speech to text. A detector scores five scam steps from the text. At 70 or more, the warning screen and family alert start.

- **User Journey**

  The user switches Nazar on and agrees to listening. A call comes in, and a risk meter and live text appear. At 70 or more the screen turns red, a voice says “Hang up”, and the family gets a Telegram message.

- **Process Flow Diagram**



Warning threshold of 70 is a starting value that we tune on test calls. Speech to text: Whisper [7].

Proposed Tech Stack

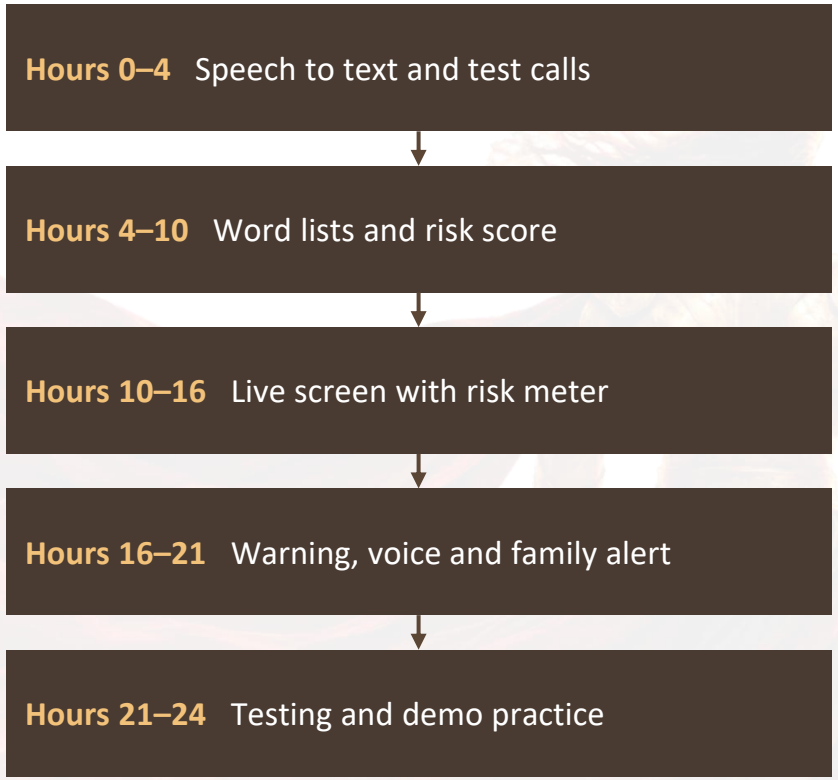
Layer	Prototype today, and what we will use in the final
Frontend	Live now: HTML, CSS and JavaScript on Vercel, with the browser's speech recognition and speech output. Final: React, TypeScript and Tailwind.
Backend	Live now: none, all scoring runs in the browser. Final: Python and FastAPI with WebSockets for live audio and live results [12], on the user's own laptop.
Database	None in the prototype, by design: session data stays in memory and is cleared when the call ends. Optional PostgreSQL later, for anonymised pilot statistics.
AI/ML Tools	Whisper-small for Hindi and Punjabi speech to text [7]; Vosk Hindi models as an alternative under evaluation, not in the prototype [8]; word lists with a scoring rule; MuRIL classifier planned [9].
APIs and Frameworks	Telegram Bot API for family alerts, working in the prototype [13]; WhatsApp click-to-chat link; pydantic and pytest for data checks and tests.

Low cost to build and pilot: every tool is free or open source and runs on a normal laptop.

Feasibility & Execution Plan

- **Development Approach**
  Done before the hackathon: the detector, 20 scripted scam and genuine test calls, and a live web prototype (next slide). In the 24 hours: on-device Whisper speech to text, Punjabi word lists, and polish.
- **Milestones**
  Five milestones across the 24 hours (see the timeline on the right).
- **Technical Challenges**
  Android blocks call recording [10][11], so the demo uses the laptop microphone. Punjabi accuracy is weaker, so the demo leads with Hindi and Hinglish. False alarms are controlled by the multi-step score.
- **Execution Strategy**
  After the hackathon: pilot with senior-citizen groups, then partner with banks and telecom operators. We will measure time from key phrase to warning, false alarms per 100 genuine calls, and scam steps caught.

24-hour timeline

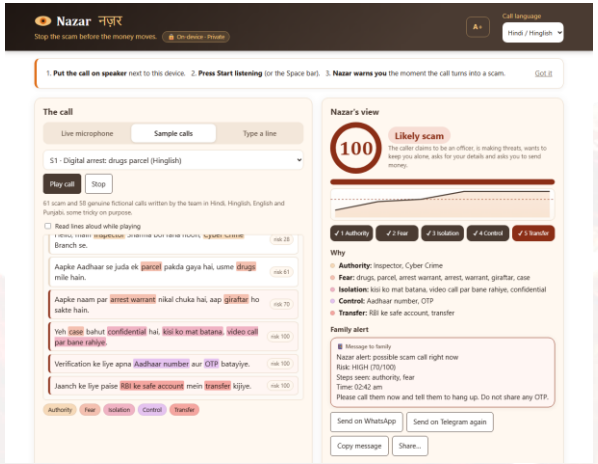


Sources: [10] Engadget and [11] The Register on Android call-recording limits.

Working Prototype

- **Live Demo**
  On-device version: start.bat, then localhost:8000 (works with Wi-Fi off). Hosted version: nazar-codevators.vercel.app. Code: github.com/Kunal039/Nazar_CodeVators.
- **What Works Today**
  Live speech to text on the device in Hindi, English and Punjabi; five-step score with highlighted reasons; spoken warning; family alert by WhatsApp or Telegram; privacy panel.
- **Results on 119 Scripted Calls**
  54 of 54 digital-arrest scams caught before money was asked for, including Whisper-style transcripts; 1 of 58 genuine calls raised the alarm (a lawyer call, documented). Through Whisper on phone-quality audio: Hindi and Punjabi scams alarm at sentence 3; the genuine call stays safe.
- **Still to Build**
  Android app with dialer integration, listening only to unknown numbers, learning weights from real call data, and pilots with banks and telecom partners.

The app today



54/54	1/58	0.7 s
scams caught	false alarms	per sentence

Try it: nazar-codevators.vercel.app

Results from node web/test.js and node web/e2e_live.js. All test calls are fictional and written by our team, so they are not real-world accuracy.

What We Built at the Venue

- **On-device Speech Server**
  A FastAPI WebSocket server runs Whisper large-v3-turbo on the laptop's GPU. The browser streams mic audio; the server cuts it into sentences and returns text in about 0.7 s each. Falls back to a small CPU model, and works with Wi-Fi off.
- **Privacy by Design (raised at Evaluation 1)**
  On-device speech by default, call text forgotten after 30 seconds, everything wiped when listening stops, and family alerts that never contain what was said.
- **Accuracy Fixes**
  Threats count only when aimed at the listener (fixed a news-report false alarm); Whisper mishearings such as ग्रग and वोरेंट added; Hinglish 'don't tell your family' phrases; delivery OTP recognised as genuine.
- **Testing**
  119 test calls (64 new), unit tests, privacy tests and an end-to-end test through Whisper, all run by GitHub Actions on every change. 230+ commits and feature pull requests on GitHub.

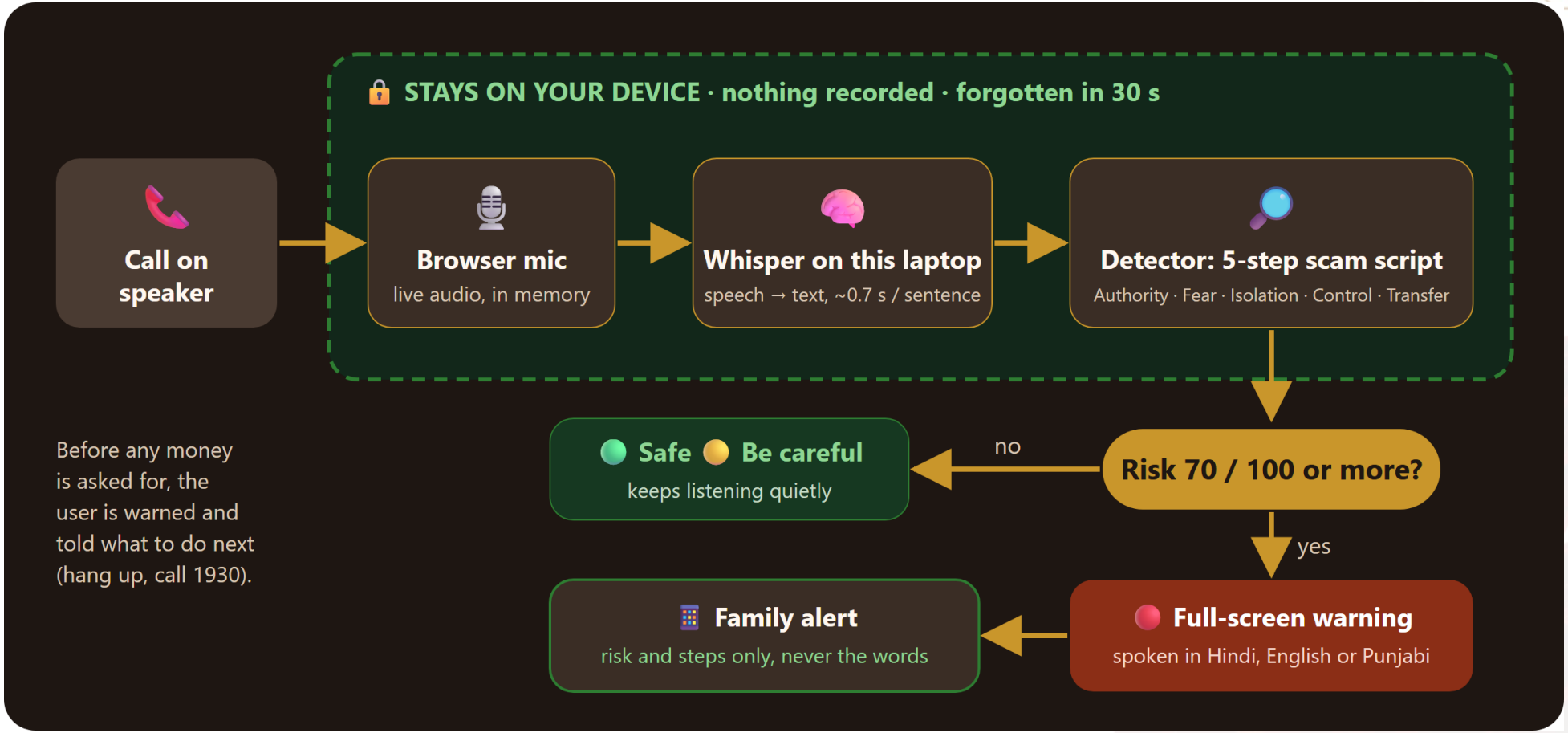
The detector and web page started as our Phase 1 online-round prototype; the GitHub README marks what was built at the venue.

Privacy by Design

- **Audio Never Leaves the Device**
  Whisper runs on the device, bound to localhost. Cloud speech is only an opt-in fallback, with a warning before it is used.
- **Nothing Is Recorded, Short Memory**
  Audio is transcribed in memory sentence by sentence and discarded. After 30 seconds a line keeps only its scam phrases, or nothing at all.
- **Wiped When the Call Ends**
  Stopping listening forgets all call text at once. No logs of call content are written anywhere.
- **Alerts Carry No Conversation**
  The family sees only the risk level, the warning signs and the time, never what was said. On a phone, Nazar would listen only to calls from unknown numbers.

A smoke detector, not a recorder. Every promise is checked by automated privacy tests on every change: PRIVACY.md on GitHub.

How It Works



Everything that touches the call stays inside the green box: on the device, nothing recorded, forgotten in 30 seconds.

Why Nazar

- **Innovation**
  Recognises the scam script itself (five steps, in order) live during the call, in Hindi, English and Punjabi, instead of blocking numbers after the damage is done.
- **Implementation**
  On-device Whisper large-v3-turbo on GPU with a FastAPI WebSocket server; rule-based detector with script folding; six test suites and privacy tests run by GitHub Actions on every change.
- **Functionality**
  Works end to end today: live mic, alarm before money is asked (54 of 54 scam calls), spoken warning, family alert, complaint report, offline, installable, and a live website.
- **Impact**
  Digital-arrest scams target older people and their savings. Nazar warns in their language, brings in family, and keeps the call private, so people can trust it on every call.

Code: github.com/Kunal039/Nazar_CodeVators · Website: nazar-codevators.vercel.app

Expected Impact & Future Scope

- **Target Impact**
  Stop scam calls before money moves, starting with senior citizens and their families. Impact will be measured in pilots by scams caught and false alarms.
- **Potential Benefits**
  Families: peace of mind and protected savings. Banks and telecom operators: fewer fraud cases and a protection feature to offer (expected, to be tested).
- **Scalability**
  Free tools on a normal laptop. The same engine can move to phones through a dialer or telecom partner [10][11]. MuRIL covers 17 Indian languages [9].
- **Future Possibilities**
  WhatsApp and video calls, SMS and link checks, one-tap report to the 1930 helpline [4], a bank link to pause risky transfers, anonymised scam trends for cyber-crime teams, a fully offline mode.

Growth path

Now
Live web prototype: nazar-codevators.vercel.app

↓

Next, 0 to 6 months
Pilots with senior-citizen groups; Android app with a partner

↓

Later, 6 to 18 months
Banks, payment apps and telecom operators

↓

Long term
More languages, offline mode, scam-trend feed

How it can earn: free for families to build trust; banks, payment apps and telecom operators pay a licence or per-user fee. Pricing and partner interest are not yet validated.

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Where a slide uses a number or claim from a source, the source number is shown in square brackets. Our own assessments and plans are marked as such on the slides.

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Thank You

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Live demo: nazar-codevators.vercel.app · Code: github.com/Kunal039/Nazar_CodeVators

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